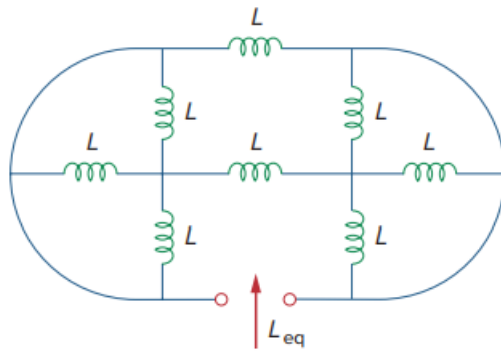


Problem

Find L_{eq} in the circuit of Fig. 6.78.

Fig. 6.78:



Step-by-step solution

Step 1 of 3

$$L \parallel (L \parallel L) = L + \frac{L \times L}{L + L}$$

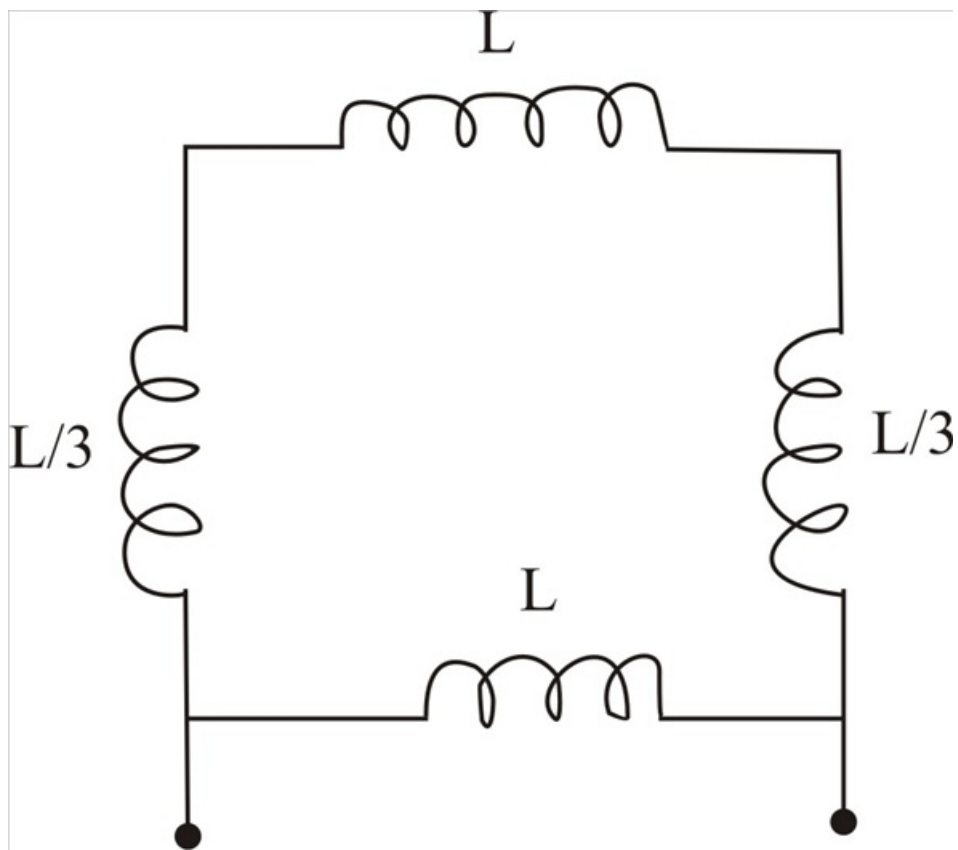
$$L \parallel (L \parallel L) = L + \frac{L}{2}$$

$$L \parallel (L \parallel L) = \frac{L \times \frac{L}{2}}{L + \frac{L}{2}}$$

$$L \parallel (L \parallel L) = \frac{L}{3}$$

Hence the given circuit is equivalent to that shown below:

Step 2 of 3



Step 3 of 3

From the above circuit,

$$L_{eq} = L \parallel \left(L + \frac{2}{3}L \right)$$

$$L_{eq} = \frac{L \times \frac{5}{3}L}{L + \frac{5}{3}L}$$

$$\therefore L_{eq} = \frac{5}{8}L$$